

Physics of raindrops

X24 (2024) — English translation

The mechanism of cloud formation can be summarized as follows. Moist air rises and cools with altitude, causing water vapor to become supersaturated, that is, its partial pressure becomes greater than the saturated vapor pressure at the corresponding temperature. In this case, water vapor begins to condense and form water droplets. As long as these droplets are small enough, they fall slowly enough and remain in the cloud. If the size of the drops is large enough that they can reach the Earth without evaporating, it begins to rain. This problem examines the mechanism of droplet formation in supersaturated water vapor and their further growth due to diffusion.

The following notation and numerical values are used in the problem: $\sigma = 7.5 \cdot 10^{-2} \text{ H/м}$ – coefficient of surface tension of water; $T = 283 \text{ K}$ – atmospheric temperature; $p_s = 1.23 \text{ кПа}$ – pressure of saturated water vapor at the considered atmospheric temperature; ρ_s – density of saturated water vapor at a given temperature; p_v, ρ_v – pressure and density of water vapor in the atmosphere; $\varphi = \rho_v/\rho_s = p_v/p_s$ – water vapor supersaturation coefficient; $\rho_L = 1.0 \cdot 10^3 \text{ kg/м}^3$ – density of water; $\mu = 0.018 \text{ kg/моль}$ – molar mass of water; m – mass of water molecules; $k = 1.38 \cdot 10^{-23} \text{ Дж/К}$ – Boltzmann constant; $L = 2.48 \text{ МДж/К}$ – specific heat of water vaporization; $R = 8.31 \text{ Дж/(моль} \cdot \text{К)}$ – universal gas constant; $N_A = 6.02 \cdot 10^{23} \text{ моль}^{-1}$ – Avogadro's constant. Water vapor can be considered an ideal gas in all parts of the problem.

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Part A. Drops in a homogeneous atmosphere (4 points)

Let the atmosphere consist only of air and water vapor without impurities. When a drop forms, additional energy must be expended to create the surface of the water. Therefore, even in the case of supersaturated water vapor, the formation of droplets is complicated by the fact that at a small size the relative value of the surface energy is large. It is known from thermodynamics that for a process at constant energy its possibility is determined by the value of the Gibbs free energy G . The greater the free energy required, the less likely the process. Let us investigate how the free energy required for the formation of a drop depends on its radius r . To do this, you will need the following facts: The free energy of a liquid surface of area A is equal to $\Delta G_{surf} = \sigma A$, σ is the coefficient of surface tension. The free energy of one mole of saturated water vapor is equal to the free energy of a mole of water (ignoring surface energy) at the same temperature and pressure. The difference between the free energy of one mole of supersaturated water vapor and saturated water vapor at the same temperature is $\Delta G_v = RT \ln \varphi$ ($\varphi = \rho_v/\rho_s > 1$ is the steam supersaturation coefficient). The free energy of water vapor is proportional to its amount of matter.

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A1 Find the change in the free energy of water vapor if a drop of radius r is formed from it. **1.00**
Express your answer in terms of $r, \sigma, \varphi, R, T, \rho_L, \mu$.

A2 Find the critical value of the droplet radius r_c at which ΔG is maximum, as well as the **0.80**
corresponding value ΔG_c . Express your answer using $\sigma, \varphi, R, T, \rho_L, \mu$. Find the numerical
value of r_c at $\varphi = 1.01$.

Until the drop has reached the radius r_c , its growth is accompanied by an increase in free energy and is therefore unlikely. As soon as the radius exceeds the critical one, further growth will occur without difficulty with a decrease in free energy. Therefore, when studying the number of droplets generated, one can focus on droplets of the critical radius. A drop can form around any of the water molecules, but the probability of such a process is small and is determined by the required free energy. From statistical mechanics it follows that the concentration of centers around which condensation can actually occur is equal to

$$n_c = n e^{-\Delta G_c / kT},$$

где n – концентрация молекул воды в паре, ΔG_c was found in the previous paragraph.

A3 Consider a drop of critical radius r_c . Determine the time τ during which the number of **0.70**
molecules in it will increase by g . Express your answer using $r_c, g, p_s, m, k, T, \varphi$. Assume
that during the growth process the radius of the drop does not change; the evaporation of
molecules from the drop can be neglected. It is known that

$$dN = dt dS \frac{p_v}{\sqrt{2\pi m k T}}$$

молекул. Здесь p_v – давление пара, m – масса молекул, T , the gas temperature, falls on
the surface area dS in time dt .

We will consider time τ to be the characteristic time of growth of droplets from the embryo. During time τ , all nuclei present in the system turn into drops of critical radius, and in their place new nuclei appear in the same quantity.

A4 Find the number of drops J that are formed per unit time in a unit volume of supersaturated **0.60**
water vapor. Express your answer in terms of $\sigma, \varphi, p_s, r_c, T, g$.

A5 From the results of the previous paragraph it follows that the rate of droplet formation very **0.90**
much depends on the steam supersaturation coefficient. Determine numerically the value of
the vapor supersaturation coefficient φ , at which one drop per second is born at a
temperature $T = 283$ K in 1 cm^3 air. Consider that $g = 100$. The remaining numerical data
are given at the beginning of the problem.

Part B. Diffusion growth of droplets (4 points)

In this part we will use the following notations (in addition to those given at the beginning of the problem): ρ_v – density of water vapor at a large distance from the drop; ρ_r – density of water vapor near the surface of the drop; ρ_s – density of saturated water vapor at atmospheric temperature at a large distance to a drop; $T = 283$ K – atmospheric temperature at a large distance to the drop; T_r – drop temperature;

$K = 2.40 \cdot 10^{-2} \text{ Дж}/(\text{м} \cdot \text{с} \cdot \text{К})$ – air thermal conductivity coefficient; $D = 2.36 \cdot 10^{-5} \text{ м}^2/\text{с}$ – diffusion coefficient of water vapor in air; r – drop radius; M – drop mass.

In this part we will use the following notations (in addition to those given at the beginning of the problem):

The drop grows due to diffusion. The rate of change in the mass of the drop and the rate of heat removal are given by the relations

$$\frac{dM}{dt} = 4\pi r D(\rho_v - \rho_r); \quad \frac{dQ}{dt} = 4\pi K r(T_r - T).$$

We assume that the temperature of the drop during its growth remains constant, and all the heat is released only due to the condensation of water.

- B1** For saturated vapor in equilibrium with a liquid, express the derivative of pressure with respect to temperature dp_s/dT in terms of p_s, L, R, T, μ . Using the result obtained, find the relative change in the density of saturated water vapor $\Delta\rho_s/\rho_s$ with a small change in temperature ΔT . Express your answer using $\Delta T, T, L, \mu, R$. You can use the relationship between small changes in pressure, density and temperature of an ideal gas **0.80**

$$\frac{\Delta p_s}{p_s} = \frac{\Delta \rho_s}{\rho_s} + \frac{\Delta T}{T}.$$

- B2** Express dQ/dt in terms of dM/dt and L . **0.20**

- B3** Using the result of the previous paragraph and the heat equation, express the temperature difference between the drop and the atmosphere, $T_r - T$, in terms of dM/dt , as well as r, L, K . **0.30**

- B4** We assume that near the surface of the drop the density of water vapor is equal to the density of saturated vapor at the temperature of the drop. Assuming the temperature and density differences are small and using the results B1, B3, express the ratio $(\rho_r - \rho_s)/\rho_s$ (ρ_r is the vapor pressure near the surface of the drop) in terms of L, r, K, μ, R, T and dM/dt . **0.30**

- B5** Using the diffusion equation, express the ratio $(\rho_r - \rho_v)/\rho_s$ in terms of $dM/dt, r, D, \rho_s$. **0.30**

- B6** By excluding the vapor density near the surface of the drop ρ_r from the answers in the two previous paragraphs, you obtain an expression for dM/dt . Express your answer in terms of $\varphi, \mu, R, T, D, p_s, L, K, r$. **0.60**

- B7** The rate of increase in the radius of the drop has the form **0.50**

$$\frac{dr}{dt} = \frac{\xi}{r^k}.$$

Определите k и ξ , выразите ответ через $\varphi, \rho_L, \mu, R, T, D, p_s, L, K$.

- B8** Find the dependence of the radius of the drop on time. The initial radius of the drop is r_0 . Express your answer in terms of r_0, ξ, t . **0.50**

B9 Let the initial radius of the drop be $r_0 = 0.7$ мкм. Find the numerical value of the time **0.50** during which it will grow to size $r_1 = 10$ мкм with a supersaturation coefficient $\varphi = 1.1$. The remaining numerical values are given at the beginning of this part.

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